Chapter 6
Eigenvalues and Eigenvectors

6.3 Diagonalization

Motivation

In this section, we consider the problem of factoring an $n \times n$ matrix A into a product of the form XDX^{-1}, where D is diagonal.

We will give a necessary and sufficient condition for the existence of such a factorization and look at some examples.

We begin by showing that eigenvectors belonging to distinct eigenvalues are linearly independent.

Theorem 6.3.1

If $\lambda_1, \lambda_2, \ldots \lambda_k$ are **distinct eigenvalues** of an $n \times n$ matrix A with corresponding eigenvectors $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k$, then the vectors $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k$ are **linearly independent**.

Warm-up questions

- Is $\mathbf{x}_1$ linearly independent?

  Yes, because eigenvectors are nonzero, and 1 nonzero vector is linearly independent.

- If $a \neq 0$, what eigenvalue does $a\mathbf{x}_1$ belong to?

  $A(a\mathbf{x}_1) = aA\mathbf{x}_1 = a\lambda_1\mathbf{x}_1 = \lambda_1(a\mathbf{x}_1)$. $\Rightarrow a\mathbf{x}_1$ also belongs to the eigenvalue λ_1.

- Is $\mathbf{x}_2$ in the span of $\mathbf{x}_1$?

  No, $\mathbf{x}_2$ belongs to eigenvalue λ_2, but any $a\mathbf{x}_1$ belongs to eigenvalue $\lambda_1 \neq \lambda_2$.

- Conclusion: $\mathbf{x}_1, \mathbf{x}_2$ are linearly independent.

Case of 3 eigenvectors

Step by step questions

We have already seen that $\mathbf{x}_1, \mathbf{x}_2$ are linearly independent.

Suppose $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$ is linearly dependent, so $\mathbf{x}_3 = a\mathbf{x}_1 + b\mathbf{x}_2$

1. Write $A\mathbf{x}_3 = \lambda_3\mathbf{x}_3$ as a linear combination of $\mathbf{x}_1, \mathbf{x}_2$

$$A\mathbf{x}_3 = \lambda_3\mathbf{x}_3 = \lambda_3 a\mathbf{x}_1 + \lambda_3 b\mathbf{x}_2$$

2. Write $A\mathbf{x}_3 = A(a\mathbf{x}_1 + b\mathbf{x}_2)$ as a linear combination of $\mathbf{x}_1, \mathbf{x}_2$

$$A\mathbf{x}_3 = A(a\mathbf{x}_1 + b\mathbf{x}_2) = a\lambda_1\mathbf{x}_1 + b\lambda_2\mathbf{x}_2$$

3. Subtract the two equations. What can you conclude as $\mathbf{x}_1, \mathbf{x}_2$ are linearly independent, and eigenvalues are distinct?

$$\mathbf{0} = a(\lambda_3 - \lambda_1)\mathbf{x}_1 + b(\lambda_3 - \lambda_2)\mathbf{x}_2$$

$$a(\lambda_3 - \lambda_1) = 0, \lambda_3 \neq \lambda_1 \Rightarrow a = 0$$

$$b(\lambda_3 - \lambda_2) = 0, \lambda_3 \neq \lambda_2 \Rightarrow b = 0$$

4. Why is this a contradiction?

$\mathbf{x}_3 = a\mathbf{x}_1 + b\mathbf{x}_2 = \mathbf{0}$, but eigenvectors must be nonzero.

5. What should the correct conclusion be?

$\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$ is also linearly independent.

Theorem 6.3.1

If $\lambda_1, \lambda_2, \ldots \lambda_k$ are **distinct eigenvalues** of an $n \times n$ matrix A with corresponding eigenvectors $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k$, then the vectors $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k$ are **linearly independent**.

Explanation

- The proof of the general case is similar (see textbook).
- Intuitive meaning: eigenvectors in distinct eigenspaces behave very differently under A, so they are linearly independent.
- If a matrix A has n distinct eigenvalues, then A has n linearly independent eigenvectors, i.e. these eigenvectors form a basis of $\mathbb{R}^n$.

Diagonalization

An $n \times n$ matrix A is said to be **diagonalizable** if there exists a nonsingular matrix X and a diagonal matrix D such that

$$X^{-1}AX = D$$

We say that X **diagonalizes** A.

What is X and D:

Matrix D: write eigenvalues of A on the diagonal:

$$D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}$$

Matrix X: write eigenvectors of A column by column:

$$X = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n]$$

Example 1

For $A = \begin{bmatrix} 2 & -3 \\ 2 & -5 \end{bmatrix}$, find X and D such that $X^{-1}AX = D$.

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Solution

Step 1: Find eigenvalues to determine D.

$$\det(A - \lambda I) = 0 \Rightarrow (2 - \lambda)(-5 - \lambda) + 6 = 0$$

$$\Rightarrow \lambda^2 + 3\lambda - 4 = 0 \Rightarrow \lambda_1 = -4\,, \lambda_2 = +1,\ D = \begin{bmatrix} -4 & 0 \\ 0 & 1 \end{bmatrix}$$

2 eigenvalues → independent eigenvectors

Step 2: Find eigenvectors to determine X and X^{-1}.

For $\lambda_1 = -4$: $6x_1 - 3x_2 = 0 \Rightarrow 2x_1 = x_2 \Rightarrow \mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

For $\lambda_2 = +1$: $x_1 - 3x_2 = 0 \Rightarrow x_1 = 3x_2 \Rightarrow \mathbf{x}_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

Let $X = [\mathbf{x}_1, \mathbf{x}_2] = \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix} \Rightarrow X^{-1} = -\frac{1}{5}\begin{bmatrix} 1 & -3 \\ -2 & 1 \end{bmatrix}$

Continued→

Example 1 (continued)

For $A = \begin{bmatrix} 2 & -3 \\ 2 & -5 \end{bmatrix}$, find X and D such that $X^{-1}AX = D$.

Checking Answer

$$D = \begin{bmatrix} -4 & 0 \\ 0 & 1 \end{bmatrix}, X = \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}, X^{-1} = \begin{bmatrix} -1/5 & 3/5 \\ 2/5 & -1/5 \end{bmatrix}$$

Calculate AX:

$$AX = \begin{bmatrix} 2 & -3 \\ 2 & -5 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} -4 & 3 \\ -8 & 1 \end{bmatrix}$$

Because columns are eigenvectors

Calculate XD:

$$XD = \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} -4 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -4 & 3 \\ -8 & 1 \end{bmatrix}$$

Because Type II **column** operations

$$AX = XD \Leftrightarrow X^{-1}AX = D$$
$$\Leftrightarrow A = XDX^{-1}$$

Theorem 6.3.2 - Diagonalization

Theorem

An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors.

Proof

- Let $X = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n], D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}. (\mathbf{x}_j \neq \mathbf{0})$
- $XD = [\lambda_1\mathbf{x}_1, \lambda_2\mathbf{x}_2, \dots, \lambda_n\mathbf{x}_n]$ (*column operations!*)
- $AX = [A\mathbf{x}_1, A\mathbf{x}_2, \dots, A\mathbf{x}_n]$ (*matrix multiplication!*)
- $AX = XD \Leftrightarrow A\mathbf{x}_j = \lambda_j\mathbf{x}_j$ for all $i \Leftrightarrow \lambda_1, \dots, \lambda_n$ are eigenvalues of A, and $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ are corresponding eigenvectors.

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Theorem 6.3.2 - Diagonalization

Theorem

An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors.

Summary from previous slide:

$D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}, X = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n], AX = XD \Leftrightarrow \mathbf{x}_j$ are eigenvectors

Therefore, A is diagonalizable

$\Leftrightarrow$ Exists nonsingular X such that $AX = XD$ (D is diagonal)

$\Leftrightarrow$ Column vectors of $X = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n]$ are linearly independent eigenvectors of A

$\Leftrightarrow$ A has n linearly independent eigenvectors $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$.

Remarks – Diagonalization

1. If A is diagonalizable, then the column vectors of the diagonalizing matrix X are **eigenvectors** of A and the diagonal elements of D are the corresponding **eigenvalues** of A.
2. The diagonalizing matrix X is **not unique**. Reordering the columns of a given diagonalizing matrix X or multiplying them by nonzero scalars will produce a new diagonalizing matrix.
3. If A is $n \times n$ and A has n **distinct eigenvalues**, then A is **diagonalizable**. If the eigenvalues are not distinct, then A may or may not be diagonalizable, depending on whether A has n linearly independent eigenvectors.
4. If A is diagonalizable, then A can be factored as a product

$$A = XDX^{-1}.$$

Diagonalization and Power

- If A is diagonalizable, then

$$X^{-1}AX = D \Leftrightarrow A = XDX^{-1}$$

(where X is nonsingular, D is diagonal)

- **Question:** Find a formula for A^k for $k = 1,2,3, \ldots$
- $A^2 = (XD\cancel{X}^{-1})(\cancel{X}DX^{-1}) = XDDX^{-1} = XD^2X^{-1}$
- In general,

$$A^k = (XDX^{-1})^k = (XD\cancel{X}^{-1})(\cancel{X}D\cancel{X}^{-1}) \ldots (\cancel{X}DX^{-1})$$

$$\Rightarrow \boxed{A^k = XD^kX^{-1}} = X\begin{bmatrix} (\lambda_1)^k & 0 & \cdots & 0 \\ 0 & (\lambda_2)^k & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & (\lambda_n)^k \end{bmatrix}X^{-1}$$

Diagonalization makes power easy to calculate.

Example 6.3.2

$A = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & -2 \\ 2 & -1 & -1 \end{bmatrix}$ is diagonalizable:

$\det(A - \lambda I) = -\lambda(\lambda - 1)^2 \Rightarrow \lambda_1 = 0, \lambda_2 = \lambda_3 = 1.$

On your own, find a basis for each eigenspace $N(A - \lambda I)$:

$\lambda_1 = 0$ with eigenvector $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$\lambda_2 = \lambda_3 = 1$ with two independent eigenvectors $\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

So $A = XDX^{-1}$, where $D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix}, X^{-1} = \begin{bmatrix} -2 & 1 & 2 \\ 1 & 0 & -1 \\ 2 & -1 & -1 \end{bmatrix}.$

1. You can check $XDX^{-1} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -2 & 1 & 2 \\ 1 & 0 & -1 \\ 2 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & -2 \\ 2 & -1 & -1 \end{bmatrix} = A$

2. Can you quickly calculate A^{100}? Hint 1: $A = XDX^{-1}$. Hint 2: What is D^{100}?

Note: Even though A only has 2 distinct eigenvalues, the matrix can still be diagonalized since there are 3 linearly independent eigenvectors.

Example 6.3.3

Is A= $\begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$ diagonalizable?

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Solution

Find eigenvalues: solve $\det(A - \lambda I) = 0$, we have:

$$\begin{bmatrix} 2-\lambda & 4 & 3 \\ -4 & -6-\lambda & -3 \\ 3 & 3 & 1-\lambda \end{bmatrix} = 0$$

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$$\Rightarrow \lambda^3 + 3\lambda^2 - 4 = 0$$

$$\Rightarrow (\lambda + 2)(\lambda + 2)(\lambda - 1) = 0$$

That is: $\lambda_1 = 1, \lambda_2 = \lambda_3 = -2$

Example 6.3.3

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Find eigenvectors:

For $\lambda_1 = 1$

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & 0 \\ -4 & -7 & -3 & 0 \\ 3 & 3 & 0 & 0 \end{array}\right] \Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] = 0$$

So, we have eigenvector $[1, -1, 1]^T$

For $\lambda_2 = \lambda_3 = -2$

$$\left[\begin{array}{ccc|c} 4 & 4 & 3 & 0 \\ -4 & -4 & -3 & 0 \\ 3 & 3 & 3 & 0 \end{array}\right] \Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] = 0$$

So, we have eigenvector $[-1, 1, 0]^T$

A does **NOT** have 3 linear independent eigenvectors

$\Rightarrow A$ is **NOT** diagonalizable.